Practical 9: Time Intelligence and Advanced DAX

Name: _____ **Roll No.:** _____ **Date:** _____

Aim

Create a date table, time-intelligence and context-modifying DAX measures; use Performance Analyzer.

Procedure

1. Create Date table; add Year, Month Number, Month Name; mark as date table; relate Date[Date] to Sales[Sales Date].

2. Create Sales YTD, Sales LY, YoY Growth %, Region Share %.

3. Matrix: Year, Month rows; Total Sales, Sales YTD, Sales LY, YoY Growth % values.

4. View > Performance Analyzer: record, note durations, remove unused columns/visuals, record again.

DAX

```
Date = CALENDAR(MIN(Sales[Sales Date]), MAX(Sales[Sales Date]))
Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY Growth % = VAR Curr = [Total Sales] VAR Prev = [Sales LY]
               RETURN DIVIDE(Curr - Prev, Prev)
Region Share % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Store[Region])))
```

Output (sample data, year boundary)

Year	Month	Total Sales	Sales YTD	Sales LY	YoY %
2024	Nov	9,025	80,330		
2024	Dec	3,755	84,085		
2025	Jan	2,189	2,189	5,501	-60.2%
2025	Feb	970	3,159	2,560	-62.1%
2025	Mar	8,103	11,262	21,446	-62.2%

Region Share %: East 34.1%, North 19.9%, South 21.7%, West 24.3% = 100%. Record your own Performance Analyzer timings (before/after).

Result: YTD resets each January; first-year LY/YoY blank; Region Share % totals 100%.

Viva Answers

Why mark a date table? Time intelligence needs a continuous date column.

CALCULATE? Evaluates an expression with a modified filter context.

VAR/RETURN? Readable, evaluated once.

ALL? Removes the Region filter from the denominator.

Paste your Power BI screenshot here: Matrix and Performance Analyzer